🚀 Lecture 04.10: Dictionary Methods & Dynamic Views

📌 Executive Architectural Summary

Dictionaries expose view objects (`keys()` , `values()` , `items()`) that reflect dynamic changes to the underlying hash table without copying keys or values into new memory lists.

🔬 Core Engineering Mechanics

1. Core Methods Suite

- `d.get(key, default)` : Retrieves value without raising `KeyError` if absent.
- `d.keys()` : Dynamic dictionary view of keys.
- `d.values()` : Dynamic view of values.
- `d.items()` : Dynamic view of `(key, value)` tuples for iteration.
- `d.update(other)` : Merges another dictionary or iterable of key-value pairs.
- `d.pop(key, default)` : Removes specified key and returns its value.

2. Dictionary Comprehensions

> $$\{k : f(k) ext{ for } k \in ext{iterable}\}$$

💻 Production Python Implementation

```
metrics = {"loss": 0.245, "accuracy": 0.945, "val_loss": 0.310}

# Iterating over items()
for metric_name, val in metrics.items():
    print(f"Metric: {metric_name:<10} = {val:.4f}")

# Dictionary Comprehension: Mapping word to its length
words = ["apple", "banana", "kiwi", "cherry", "mango"]
length_map = {w: len(w) for w in words}
print("
```

```
Word Length Map Comprehension:")
print(length_map)
```